

CHAPTER TWO: Statistical Estimation

Introduction

It is the procedure of using a sample statistic to estimate a population parameter. This is one way of making inference about the population parameter where the investigator does not have any prior notion about values or characteristics of the population parameter. A statistic used to estimate a parameter is called *an estimator* and the value taken by the estimator is called *an estimate*. Statistical estimation is divided into two main categories: Point Estimation and Interval Estimation.

2.1 Point Estimation

When we use a single value of a statistic to estimate the corresponding parameter of a population, it is called *point estimation*. It is a common way of estimating a parameter, where a random sample of n observations is selected from a population and the statistic is calculated.

Examples:

- A sample mean is an estimate for population mean μ . That is, $\bar{X}$ is an estimator for population mean μ .
- A sample variance is an estimate for population variance. That is, S^2 is an estimator for population Variance σ^2 .
- A sample proportion estimate for population proportion.

2.1,1 Point estimation of the mean and proportion

Hence, $\bar{x} = \frac{\sum f_i}{n} = \frac{\sum x_i}{n}$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{s^2}$$

$$\hat{p} = \frac{x}{n}$$

Where, p = proportion mean

x = is the part of the sample

E.g.1: A simple random sample of 5 months of sales data provided the following information.

Month	1	2	3	4	5
Units sold	94	100	85	94	92

Thus,

- Find the point estimate of the population mean under units sold per month.
- Develop a point estimate of the population standard deviation σ .

Solution a,

$$\bar{x} = \frac{\sum x_i}{n} = \frac{94 + 100 + 85 + 94 + 92}{5} = \frac{465}{5} = \underline{93}$$

Thus, 93 units is the point estimation of μ .

Solution b, $s = \sqrt{s^2}$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} =$$

$$(94-93)^2 + (100-93)^2 + (85-93)^2 + (94-93)^2 + (92-93)^2 = 1+49+64+1+1 = \mathbf{116}$$

$$\frac{116}{5-1} = \frac{116}{4} = \mathbf{29}$$

$$s^2 = \mathbf{29}$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{29} = \mathbf{5.38}$$

Then, 5.38 units is the point estimation of σ .

E.g. 2: A survey question for the sample of 150 individuals yielded 75 yes response, 55 o response and 20 no opinion.

A. what is the point estimate of the proportion in the population who response yes?

B. what is the point estimate of the proportion in the population who response no?

C. what is the point estimate of the proportion in the population who response no opinion?

Solution a.

$$n = 150$$

$$X \text{ (the no of yes response)} = \mathbf{75}$$

$$\bar{p} = \frac{x}{n} = \frac{75}{150} = \mathbf{0.5}$$

Solution b.

$$n = 150$$

$$X \text{ (the no of no response)} = \mathbf{55}$$

$$\bar{p} = \frac{x}{n} = \frac{55}{150} = \mathbf{0.367}$$

Solution c.

$$n = 150$$

$$X \text{ (the no of no opinion response)} = \mathbf{20}$$

$$\bar{p} = \frac{x}{n} = \frac{20}{150} = \mathbf{0.133}$$

2.2 Interval estimation

It is unlikely that any particular estimate will be exactly equal to the population mean, surely an estimate can be greater than or less than the parameter. That is, it is not always possible to estimate population parameter without any error so allowance is needed for such error. We take interval, ranges of values about an estimate in which the parameter may lie. This procedure is Interval estimation.

It is the procedure that results in the interval of values of a parameter. Interval estimates indicate the precision or accuracy of an estimate and are, therefore, preferable to point estimates. It deals with identifying the upper and lower limits of a parameter. Confidence interval for the parameter is:

Estimate $\pm$ critical value $\times$ Standard error of the estimator

Example3: Confidence interval for the population mean is:

$\bar{X} \pm \text{Critical value} \times \text{Standard error of } (\bar{X})$

2.2.1 Confidence interval Estimation for population means

Although $\bar{X}$ possesses nearly all the qualities of a good estimator, because of sampling error, we know that it's not likely that our sample statistic will be equal to the population parameter, but instead will fall into an interval of values. We will have to be satisfied knowing that the statistic is "close to" the parameter. That leads to the obvious question, what is "close"?

We can phrase the latter question differently: How **confident** can we be that the value of the statistic falls within a certain "distance" of the parameter? Or, what is the probability that the parameter's value is within a certain range of the statistic's value? This range is the *confidence interval*.

The *confidence level* is the *probability* that the value of the parameter falls within the range specified by the confidence interval surrounding the statistic. There are different cases to be considered to construct confidence intervals.

Case-I: *Population standard deviation (σ) is known.*

$$(\bar{X} - Z_{\alpha/2} \sigma / \sqrt{n}, \bar{X} + Z_{\alpha/2} \sigma / \sqrt{n}).$$

This interval is known as a $(1 - \alpha)$ 100% confidence interval for population mean (μ).

Here are the Z values corresponding to the most commonly used confidence levels.

$(1 - \alpha) 100\%$	α	$\alpha/2$	$Z_{\alpha/2}$
90	0.10	0.05	1.645
95	0.05	0.025	1.96
99	0.01	0.005	2.58

Example 4: The weights of full boxes of a certain kind of cereal are normally distributed with a standard deviation of 0.27 ounce. If a sample of 15 randomly selected boxes produced a mean weight of 9.87 ounce, find:

- The 95% confidence interval for the true mean weight of boxes of this cereal,
- The 99% confidence interval for the true mean weight of boxes of this cereal,
- What effect does the increase in the level of confidence have on the width of the interval?

Solution:

a) Given $r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$, so that $r = \frac{\sum xy - n\bar{x}\bar{y}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}}$, $r = \frac{\sum xy - n\bar{x}\bar{y}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}}$.

The 95% C.I. is $\frac{520 - (25)(5)(4)}{\sqrt{(650 - 25(5)(5))(436 - (25)(4)(4))}}$ and r_s ounce

Where r_s

Substituting these values in $r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$, the resulting confidence interval is (9.73, 10.01).

- Similarly the 99% C.I. is (9.69, 10.05).
- The increase in the confidence level widens the length of the confidence interval.

Case-II: when population standard deviation (σ) is unknown and the sample size is large enough. A $(1-\alpha) 100\%$ confidence interval for population mean (μ) is

$$(\bar{X} - Z_{\alpha/2} \sigma / \sqrt{n}, \bar{X} + Z_{\alpha/2} \sigma / \sqrt{n}), \text{ if } \sigma \text{ known}$$

$$(\bar{X} - Z_{\alpha/2} S / \sqrt{n}, \bar{X} + Z_{\alpha/2} S / \sqrt{n}), \text{ if } \sigma \text{ unknown}$$

Example 5: An economist wants to estimate the average amount in checking accounts at banks in given region. A random sample of 100 accounts gives $\bar{X} = \$357.60$ and $S = \$140.00$. Give a 95% confidence interval for μ , the average amount in any checking account at a bank in the given region.

Solution:

Given: $n = 100$, $\bar{X} = \$357.60$, $S = \$140.00$ & $\alpha = 0.05$

A 95% confidence interval for population mean (μ) is

$$\begin{aligned} & (\bar{X} - Z_{\alpha/2} S / \sqrt{n}, \bar{X} + Z_{\alpha/2} S / \sqrt{n}) \\ & = (357.60 - 1.96(140.00 / \sqrt{100}), 357.60 + 1.96(140.00 / \sqrt{100})) \\ & = (330.16, 385.04) \end{aligned}$$

Case-III: When sampling is from normally distributed population with unknown population standard deviation and when the sample size is small ($n < 30$).

When population standard deviation σ is unknown, we estimate it by sample standard deviation.

the standardized distributions of the sample mean, $t = \frac{\bar{X} - \mu}{S / \sqrt{n}}$ is t-distribution with $(n-1)$ degrees of freedom. From this distribution, $(1-\alpha)$ 100% confidence interval for population mean is

$$(\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}).$$

Example 6: From a normal sample of size 25 a mean of 32 was found .Given that the standard deviation is 4.2. Find

a) A 95% confidence interval for the population mean.

b) A 99% confidence interval for the population mean.

Solution: a/ Given: $n = 25$, $\bar{X} = 32$, $S = 4.2$, $1-\alpha = 0.95$ $\alpha = 0.05$, $\frac{\alpha}{2} = 0.025$

$$t_{0.025, 24} = 2.064 \text{ from table.}$$

$$\begin{aligned} & \text{The required interval will be } (\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}}) \\ & = 32 \pm t_{0.025, 24} \times \frac{4.2}{\sqrt{25}} \end{aligned}$$

$$\begin{aligned}
&= 32 \pm 2.064 \times \frac{4.2}{\sqrt{25}} \\
&= 32 \pm 1.73 \\
&= \underline{(30.27, 33.73)}
\end{aligned}$$

b/ Given: $n = 25$, $\bar{X} = 32$, $S = 4.2$, $1 - \alpha = 0.99$, $\alpha = 0.01$, $\frac{\alpha}{2} = 0.005$

$$t_{0.005, 24} = 2.797 \text{ i table.}$$

The required interval will be $(\bar{X} - t_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2} \frac{S}{\sqrt{n}})$

$$\begin{aligned}
&= 32 \pm t_{0.005, 24} \times \frac{4.2}{\sqrt{25}} \\
&= 32 \pm 2.797 \times \frac{4.2}{\sqrt{25}} \\
&= 32 \pm 1.35 \\
&= \underline{(29.65, 34.35)}
\end{aligned}$$

2.2.2 Sample size determination in estimation of population mean

In the process of estimating population mean μ using the sample mean with absolute margin of error (d) and risk probability α , the sample size is given by:

$$n = \left[\frac{Z_{\frac{\alpha}{2}} \sigma}{d} \right]^2 \text{ Where } |\bar{X} - \mu| = d$$

Example 7: To determine the average amount of time students take to get from one class to the next, how large a sample is needed with probability 0.95 that the error will be at most 0.25 minutes, if r_s is known from past experience to be 1.50 minutes?

Solution: Using $x's$, and replacing $y's$, and r_s in the formula for r_s , we get

$$r_s = 1 - \frac{6(20)}{10(100-1)} = 0.88 \text{ (always rounded to the next integer) is required for the estimate.}$$

2.2.3 Confidence interval for population proportion

The confidence interval for the population proportion is performed in the same manner as the population mean. We have discussed that the sampling distribution of sample proportion is

normal. The sample estimate of population proportion P is sample proportion p and sample estimate of variance of sample proportion is $Var(\hat{p}) = \frac{pq}{n-1}$ for large sample $Var(\hat{p}) = \frac{pq}{n}$.

A $(1-\alpha)$ 100% confidence interval for proportion p is given by (for large n):

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Example 8: The Human Resource director of a large organization wanted to know what proportion of all persons who had ever been interviewed for a job with his organization had been hired. He was willing to settle for 95% confidence interval. A random sample of 500 interview records revealed that 76 or 0.152 of the persons in the sample had been hired.

Solution:

Given: $\hat{p} = 0.152$, $\hat{q} = 1 - \hat{p} = 0.848$, $n = 500$, $\alpha = 0.05$, $Z_{0.025} = 1.96$

The 95% confidence interval for the population proportion is given by

$$\begin{aligned} \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} &= 0.152 \pm 1.96 \sqrt{\frac{0.152 \times 0.848}{500}} = 0.152 \pm 0.031 \\ &= \underline{(0.121, 0.183)} \end{aligned}$$

Hence the required proportion varies between 0.121 and 0.183.